

The Mean-Field Dynamics of Transformers

Motivation and description of the research paper by Philippe Rigollet

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Outline

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Why study Transformer dynamics?

- Transformers dominate modern large language models, yet their **internal behavior during forward propagation** remains poorly understood.
- A rigorous account of how token representations evolve across layers has direct practical value for:
 - designing **deeper** architectures,
 - improving **expressivity**, and
 - ensuring **stability**.

Representation collapse

The problem: Token embeddings progressively converge toward a single point as depth increases.

Consequences:

- Degrades expressivity.
- Limits the benefit of additional layers.

Key questions:

- What are the mathematical origins of collapse?
- Which design choices accelerate or delay it? (e.g. **Pre-LN vs. Post-LN**)

Why the mean-field framework?

Placing tokens on the unit sphere and treating attention as a kernel-driven force between particles unlocks:

- tools from **mean-field theory**,
- **Wasserstein gradient flows**,
- synchronization models such as the **Kuramoto model**.

Takeaway

This framework offers analytical depth that purely empirical approaches cannot provide.

The core framework

Rigollet recasts Transformer attention as an **interacting particle system** on the unit sphere:

- Each token $\leftrightarrow$ a particle on $\mathbb{S}^{d-1}$.
- Layer normalization $\leftrightarrow$ projection onto $\mathbb{S}^{d-1}$.
- Attention kernel:

$$K(x, y) = e^{\beta \langle x, y \rangle}$$

acts as an **attractive force** between tokens.

- Discrete layer updates $\leftrightarrow$ continuous-time flow.

This places Transformers within the theory of **Neural ODEs** and **mean-field dynamics**.

Main theoretical results

- **Global Clustering Theorem:** tokens asymptotically collapse to a single point.
- **Metastable regime:** multiple clusters persist before merging through *saddle-to-saddle* transitions.
- **Equiangular reduction:** a tractable model yielding *exact* clustering rates.
 - Post-LN $\Rightarrow$ **exponential** contraction.
 - Pre-LN $\Rightarrow$ **polynomial** contraction.
- **Phase transition** in long-context attention under logarithmic scaling $\beta_n = \gamma \log n$:
 - uniform averaging,
 - critical sparse mixing,
 - effective attention suppression.

How the kernel works

When particle i updates its position, it computes a weighted average of all other particles:

$$\dot{x}_i(t) \propto \sum_{j=1}^n K(x_i, x_j) x_j$$

The kernel acts as a **weight**:

- $K(x_i, x_j)$ measures how much particle j “pulls” on particle i .
- Larger kernel value $\Rightarrow$ stronger influence.
- Normalization of weights yields a probability distribution (the **softmax**).

McKean–Vlasov type

Particles interact through their *collective distribution*, creating a self-reinforcing feedback loop.

The Transformer Party: the participants

View the architecture not as a black box of numbers, but as a **dynamic social system**.

Tokens as particles: each token (a word or image patch) is a **person** in a large, circular room.

- **The room:** the surface of the unit sphere $\mathbb{S}^{d-1}$. Everyone is equidistant from the center but free to move around the perimeter.
- **Position:** a person's location represents their “meaning” or “identity” (x_i).

Attention as socializing

In a Transformer, every token “looks” at every other token — the act of **knowing each other**.

- **Similarity:** two people measure what they have in common via the inner product $\langle x, y \rangle$.
- **The pull:** recognizing similarities creates a “social pull” to move closer.
- In the paper, this is the **Unnormalized Self-Attention (USA)** update.

Energy minimization = reducing social tension

The paper defines an energy function $\mathcal{E}_\beta(\mu)$. In our analogy: **social tension**.

$$\mathcal{E}_\beta(\mu) = \frac{1}{2\beta} \iint e^{\beta \langle x, y \rangle} d\mu(x) d\mu(y)$$

- **The goal:** the system naturally minimizes tension.
- **The result:** tension is highest when similar tokens are far apart; as they move closer, energy drops.

Metastability = formation of cliques

People don't merge into one giant crowd immediately — instead, they form **cliques**.

- **The plateau:** small groups of like-minded people form quickly. Inside a tight circle, they are content $\Rightarrow$ a **metastable state**.
- **The staircase:** social tension stays flat while cliques talk amongst themselves; it drops again only when two cliques realize they have something in common and merge.

Representation collapse: the final convergence

If the party runs too long (too many layers), the **Global Clustering Theorem** kicks in:

- Eventually every person merges into a single spot.
- The pairwise “distance” between everyone is zero.
- Energy is minimized, but the party is over: **no diversity remains**.

Lesson

Minimum energy is not the same as useful representation. Collapse is the *price* of unrestrained attraction.

The perfectly symmetric party

Question: what if every guest starts at *exactly the same social distance* from every other guest?

$$\langle x_i(0), x_j(0) \rangle = \rho_0, \quad \text{for all } i \neq j.$$

The setup: a room of polite strangers, perfectly symmetric. Every pair shares the same “familiarity score” ρ .

Why symmetry helps

Key insight: because everyone is interchangeable, **symmetry never breaks**.

- At every time t , all pairs share the same $\rho(t)$.
- The high-dimensional system collapses to a **single scalar ODE**.

Why this is useful:

- No more multi-body dynamics — just one variable $\rho(t)$.
- We can compute *exactly* how fast the system converges toward full clustering ($\rho \rightarrow 1$).

Pre-LN vs. Post-LN: the contraction rates

The equiangular lens makes the normalization distinction transparent:

- **Post-LN:** $1 - \rho(t) \sim e^{-2t}$ (exponential rush to collapse)
- **Pre-LN:** $1 - \rho(t) \sim 1/t^2$ (slow polynomial drift)

Party interpretation:

- **Post-LN** = pressure-cooker; everyone bonds almost instantly.
- **Pre-LN** = slow mixer; meaningful structure has time to emerge across depth.

Why the toy model is faithful to real Transformers

In high dimensions ($d \rightarrow \infty$), randomly initialized tokens are nearly orthogonal:

$$\rho_0 \approx 0.$$

- The equiangular assumption **approximately holds at initialization** in practice.
- The model predicts real deep-Transformer behavior remarkably well.
- Confirmed by the phase-transition diagrams in Figure 3 of the paper.

Conclusion

- The paper challenges the view of AI as a black box.
- Using **particle dynamics**, it explains what happens at each layer of the Transformer.
- A very promising direction.

Future directions

What other dynamics could serve as an even more faithful abstraction of the self-attention mechanism?

Thank you!

Questions?